



Name: _____

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Learning Objectives

- Factor using GCF, difference of squares, and perfect square trinomials
- Factor trinomials of the form $ax^2 + bx + c$ by grouping
- Factor sums and differences of cubes
- Perform polynomial long division

Always check for a GCF first before applying other factoring techniques. For trinomials, use the ac-method (factor by grouping) when the leading coefficient $\neq 1$.

1. Factor out the Greatest Common Factor (GCF).

$$12x^3 - 8x^2 + 4x$$

Answer: _____

2. Factor the difference of squares.

$$25x^2 - 49$$

Answer: _____

3. Factor the perfect square trinomial.

$$x^2 + 10x + 25$$

Answer: _____

4. Factor the trinomial.

$$x^2 - 3x - 18$$

Answer: _____

5. Factor by grouping.

$$3x^3 + x^2 - 12x - 4$$

Answer: _____

6. Factor the trinomial with leading coefficient $\neq 1$.

$$2x^2 + 7x + 3$$

Answer: _____

7. Factor the sum of cubes.

$$x^3 + 27$$

Answer: _____



8. Factor completely.

$$3x^4 - 75x^2$$

Answer: _____

9. Factor the difference of cubes.

$$8x^3 - 125$$

Answer: _____

10. Perform polynomial long division.

$$(x^3 - 2x^2 + 4x - 8) \div (x - 2)$$

Answer: _____





Problems 1–4: build from simplest to more complex factoring. Problem 5 (grouping): connect to synthetic division. Problem 10 (long division): preview of rational root theorem.

Solutions

1. Factor out the Greatest Common Factor (GCF).

$$12x^3 - 8x^2 + 4x$$

→ GCF of 12, 8, 4 = 4. GCF of x^3 , x^2 , $x = x$. GCF = $4x$.

→ $4x(3x^2 - 2x + 1)$.

Answer: $4x(3x^2 - 2x + 1)$

2. Factor the difference of squares.

$$25x^2 - 49$$

→ Pattern: $a^2 - b^2 = (a - b)(a + b)$. Here $a = 5x$, $b = 7$.

→ $(5x - 7)(5x + 7)$.

Answer: $(5x - 7)(5x + 7)$

3. Factor the perfect square trinomial.

$$x^2 + 10x + 25$$

→ Recognize: $a^2 + 2ab + b^2 = (a + b)^2$ with $a = x$, $b = 5$.

→ Check: $2(x)(5) = 10x$ ✓. Answer: $(x + 5)^2$.

Answer: $(x + 5)^2$

4. Factor the trinomial.

$$x^2 - 3x - 18$$

→ Find two numbers: multiply to -18 , add to -3 → -6 and $+3$.

→ $(x - 6)(x + 3)$.

Answer: $(x - 6)(x + 3)$

5. Factor by grouping.

$$3x^3 + x^2 - 12x - 4$$

→ Group: $(3x^3 + x^2) + (-12x - 4)$.

→ $x^2(3x + 1) - 4(3x + 1) = (3x + 1)(x^2 - 4)$.

→ Factor further: $(3x + 1)(x - 2)(x + 2)$.

Answer: $(3x + 1)(x^2 - 4) = (3x + 1)(x - 2)(x + 2)$



6. Factor the trinomial with leading coefficient $\neq 1$.

$$2x^2 + 7x + 3$$

→ $a \cdot c = 6$. Find factors of 6 that add to 7: 1 and 6.

→ Split: $2x^2 + x + 6x + 3 = x(2x+1) + 3(2x+1) = (2x+1)(x+3)$.

Answer: $(2x + 1)(x + 3)$

7. Factor the sum of cubes.

$$x^3 + 27$$

→ Pattern: $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$. $a=x$, $b=3$.

→ $(x+3)(x^2 - 3x + 9)$.

Answer: $(x + 3)(x^2 - 3x + 9)$

8. Factor completely.

$$3x^4 - 75x^2$$

→ GCF = $3x^2$: $3x^2(x^2 - 25)$.

→ $x^2 - 25$ is difference of squares: $(x-5)(x+5)$.

→ Complete factorization: $3x^2(x-5)(x+5)$.

Answer: $3x^2(x - 5)(x + 5)$

9. Factor the difference of cubes.

$$8x^3 - 125$$

→ Pattern: $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$. $a=2x$, $b=5$.

→ $(2x-5)((2x)^2 + (2x)(5) + 5^2) = (2x-5)(4x^2 + 10x + 25)$.

Answer: $(2x - 5)(4x^2 + 10x + 25)$

10. Perform polynomial long division.

$$(x^3 - 2x^2 + 4x - 8) \div (x - 2)$$

→ $x^3 \div x = x^2$. $x^2 \cdot (x-2) = x^3 - 2x^2$. Subtract: $-2x^2 + 2x^2 = 0$.

→ Bring down: $4x - 8$. $4x \div x = 4$. $4 \cdot (x-2) = 4x - 8$. Remainder = 0.

→ Quotient: $x^2 + 4$.

Answer: $x^2 + 4$, remainder 0

